Homework

DM6 - Study of a logarithmic function - 55 min.

Consider the function f defined on $]0;+\infty[$ by $f(x)=\frac{1+\ln x}{x}$. Its graph is denoted $\mathcal{C}_f$.

1. Determine the limits of f at the endpoints of the domain, study its sign, then draw up its table of variations.
2. Study the convexity of f and determine the coordinates of the inflection point of $\mathcal{C}_f$.
3. Prove that the equation $f(x)=\frac{1}{2}$ has exactly two solutions. For each, give a pair of bounds justified using calculated values.
4. Using the maximum of f, prove that $1+\ln x \leq x$ for all $x>0$. State the case of equality.